What Does Spiking Cost an Autoencoder? A Leakage-Controlled Comparison with a Matched Non-Spiking Control on Speech and EEG

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***Abstract*—Spiking autoencoders are usually compared with non-spiking networks that differ from them in more than the neuron, and mostly on images. We ask how much reconstruction quality the spike costs, and what it returns in energy, for short speech and single-channel EEG windows. The spiking autoencoder (SNN-AE) is compared with a matched control (ANN-AE) that copies its selected widths, initializer and training and replaces only the leaky integrate-and-fire neuron by a leaky ReLU, and with PCA. The SNN-AE selects its architecture, the number of simulation steps, the initial threshold and decay, the spike encoding and an input transform by multi-objective search inside nested leave-one-group-out cross-validation on four datasets, and differences are tested at the recording level. The target is a low-power device, so energy is the cost we put first: it is estimated per window with a published analytical model that counts memory accesses and address arithmetic as well as operations, at published 45 nm prices, and every energy comparison is repeated under each of three memory sizes, with arithmetic alone, and with a direct-coded input's first layer recomputed at every step. Searched LSTM, GRU and Transformer autoencoders, which differ from the SNN-AE in more than the neuron, were measured once, under one reduced search budget on development folds, for their training and inference cost and their quality, and left out of the main comparison. A design study on a development fold, excluded from the headline numbers, shows that two avoidable choices – scoring every simulation step and a spiking decoder – enlarged the gap, that the initializer decided the EEG comparison (with fan-in scaling in both arms the control was handicapped and the gap vanished), and that what remains is reproduced by a non-spiking network with a hard-clipped activation: the cost is the saturation of the neuron's transfer function, not the spike.**

***Index Terms*—spiking neural networks, autoencoder, matched control, energy estimate, nested cross-validation, speech, EEG**

I. Introduction

Spiking neural networks (SNNs) communicate with binary events, so a synapse whose input did not spike does no work, and a synapse whose input did spike adds a weight instead of multiplying by it [1], [2]. Surrogate gradients [3]–[5] made such networks trainable end to end, and autoencoders are an obvious use: compress a signal window to a small code on a low-power device and reconstruct it elsewhere.

Whether that trade is worth it is a question about two numbers measured on the *same* network: how much reconstruction error the spike adds, and how much energy it saves. The second must count what spiking adds as well as what it removes: a spike turns a multiply–accumulate into an add, but every operand must also be fetched, at a higher price than the arithmetic, and a spiking layer reads and writes its membrane potentials at every simulation step [6], [7]. Published comparisons answer a different question. Spiking autoencoders have been reported as comparable to conventional ones on MNIST-type images, against baselines of similar size [8] or the same architecture [9]; when the comparison is against recurrent or attention models, those differ in recurrence, sequence length and capacity at once, so a quality difference cannot be attributed to the neuron. We are not aware of a matched comparison for the reconstruction of speech or EEG.

Two further confounds are easy to miss. First, evaluation: windows of one recording and one speaker are near-duplicates, so pooling windows before splitting measures memorization, and selecting hyperparameters on the test speaker biases every number [10]. Second, the comparison can be skewed by design choices that are not the spike, and in our case were: Section V documents a per-step loss and a spiking decoder that inflated the gap, and an initializer that, applied to both arms, handicapped the control enough to make the gap vanish on EEG.

This paper makes four contributions.

1) A *matched non-spiking control*: the ANN-AE copies the selected SNN-AE's widths, initializer, optimizer and data, and differs from it only in the encoder's hidden neuron, so the gap between them is the cost of spiking.
2) A leakage-controlled protocol: nested 6-fold leave-one-group-out cross-validation on two speech and two EEG datasets, with the SNN-AE's architecture selected on inner validation groups only, and recording-level paired statistics.
3) A quality–energy accounting that counts memory accesses and address arithmetic as well as operations, with the analytical model of [11] at the 45 nm prices of [6], applied unchanged to every model, the SNN's counts taken from its executed forward pass, and every comparison checked under three memory sizes, under arithmetic alone, and with a repeated input recomputed at every step.
4) A design study, on a development fold kept out of the headline numbers, that locates where the SNN-AE's reconstruction error comes from, and a pre-validation on the same folds that measures what searched sequence

autoencoders cost to train and to run.

II. Related Work

Spiking autoencoders. Comşa et al. [8] trained spiking autoencoders with temporal coding and found reconstruction errors comparable to conventional autoencoders of similar size on MNIST and Fashion-MNIST. The fully spiking VAE [9] compared against a non-spiking VAE of the same architecture on image datasets. Both support the matched design we adopt; neither addresses temporal signals or a group-disjoint protocol.

Training and coding. We train with surrogate gradients and backpropagation through time [3], [12], [13], and use the three common input codes: direct currents, rate (Bernoulli) coding, and latency coding [14], [15].

Energy accounting. A widespread convention prices an SNN's analog-input layers by multiply–accumulates (MACs) and its spike-driven layers by accumulations (ACs), with 45 nm figures from Horowitz [6]; Spikformer [16] is one example. It leaves out memory, which in the same table costs more than the arithmetic it feeds (a 32-bit float add 0.9 pJ, one access to a 32 KB SRAM 20 pJ), and with it the traffic only an SNN has: its potentials, read and written at every step. Estimates that count memory do not agree on the outcome, which depends on the spike rate, the number of steps and the hardware: Lemaire et al. [11] find their SNNs 6 to 8 times more efficient than non-spiking networks, Dampfhoffer et al. [7] that SNNs match efficient ANN accelerators only at very high sparsity, and Yan et al. [17] that an advantage requires strict limits on both. We therefore count memory, and report whether each conclusion survives a change of the memory model (Section III-G).

Sequence autoencoders and protocol. LSTM [18], [19], GRU [20], [21] and self-attention [22], [23] autoencoders are the usual non-spiking baselines for sequences; because they differ from an SNN-AE in more than the neuron, we measure them in a pre-validation only (Section V-A). Group-disjoint, nested evaluation is the standard remedy for leakage in model selection [10].

III. Methods

A. Task

A window $x \in \mathbb{R}^M$ of $M = 256$ consecutive samples is z-scored on its own statistics and reconstructed through a latent $z \in \mathbb{R}^d$, minimizing the mean squared error. The latent width d is fixed per signal domain ($d = 16$ for speech, $d = 64$ for EEG) and is never searched, so every model faces the same compression ratio. A recording's last window is zero-padded; padded samples are excluded from normalization, every loss and every reported error.

B. SNN-AE

Neuron. The encoder's hidden layers are leaky integrate-and-fire (LIF) neurons [24] simulated for T steps:

$$V[t] = \beta\, V[t-1] + I[t], \qquad s[t] = \mathbf{1}[V[t] > V_{th}], \qquad (1)$$

and $V[t] \leftarrow 0$ after a spike (hard reset), with $\beta = e^{-\Delta t/(RC)}$ and $\Delta t = 1$. R, C and V_{th} are trained per layer (clamped above 10^{-6}) from initial values $R = 1$, $C = -1/\ln \beta^{(0)}$ and $V_{th}^{(0)}$, so the searched $\beta^{(0)}$ is the initial decay. The spike's zero gradient is replaced in the backward pass by the arctan surrogate [12] $\hat{\sigma}'(V) = \alpha/\big(2(1 + (\frac{\pi}{2}\alpha(V - V_{th}))^2)\big)$, $\alpha = 2$, and the error is propagated through time as

$$\delta_l[t] = \hat{\sigma}'(V_l[t]) \odot \big(W_{l+1}^{\top}\delta_{l+1}[t]\big) + \beta\, \delta_l[t+1]. \qquad (2)$$

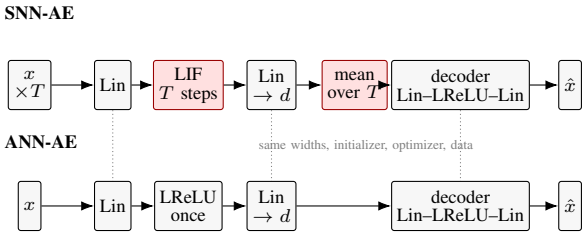


Figure 1. The SNN-AE and its matched control. Shared blocks have identical widths, initializer and training; only the encoder's hidden neuron differs (red: the parts that exist only in the spiking network).

Network. The encoder is $\mathrm{Lin}(M{\to}w_1) \to \mathrm{LIF} \to \cdots \to \mathrm{Lin}(w_{L-1}{\to}d)$ with $w_1 > \cdots > w_{L-1} > d$; the latent layer is analog. The latent is averaged over the T steps and decoded *once* by the mirror image of the encoder with leaky ReLUs (slope 0.01) and a linear output (Fig. 1). The loss is computed on that single reconstruction. Section V shows why: the earlier design, which scored the reconstruction of every step and spiked in the decoder too, was measurably worse for reasons unrelated to spiking.

Input. The window is presented for T steps as a constant current (*direct*), as Bernoulli spikes with probability proportional to the min–max-scaled sample (*Poisson*), or as one spike per sample at a time that decreases with its value (*latency*). An optional input transform is applied before the network: none (*dense*), a fixed 3-tap smoothing along the window (*conv1d*), or one LIF cell per sample with the network's threshold and decay (*recurrent*), which turns any code into spikes.

C. Matched control (ANN-AE)

For every dataset, fold and run, the ANN-AE takes the widths of that run's selected SNN-AE and replaces each hidden LIF by a leaky ReLU applied once to the analog window; it is trained with the same initializer, optimizer, data and stopping rule. It is *not* searched. The SNN-AE therefore receives the larger search budget, which favours it, and a gap in the ANN-AE's favour is conservative.

D. References

PCA with $k = d$ components, fitted on all non-test windows of the fold, is the best linear code through the same bottleneck; a mean-window predictor fixes the scale of "no model".

E. Architecture search

The SNN-AE's architecture is selected by NSGA-II [25] with population 8 for 5 generations (48 evaluations), minimizing validation MSE and an analytic inference cost. The final front is re-scored over 3 seeds and the member with the lowest

re-scored validation MSE wins (ties: lower cost). Genes: depth $L \in [1, 4]$ and widths in $(d, 128]$, $T \in [2, 32]$ (an integer drawn uniformly), $V_{th}^{(0)} \in [0.5, 1.5]$, $\beta^{(0)} \in [0.8, 0.99]$, encoding and input transform. The cost objective is a spike-blind MAC count in which T multiplies every encoder layer; it keeps cheaper genomes on the front, but the winner is chosen by validation MSE alone, so a longer simulation is selected only when it validates better.

F. Training

All models use Adam [26] (learning rate 10^{-3}; 10^{-4} for R, C, V_{th}), batch size 1 and at most 30 epochs. Early stopping [27] has patience 5, counted only once the monitored loss is 10% below that of the mean predictor (the mean training window, scored on the same monitor windows). On z-scored windows every network starts on that plateau, and in an earlier run of the pre-validation (Section V-A) plain patience stopped networks that had not left it yet, which then entered the comparison as the mean predictor under another name. A run that never reaches that level trains all 30 epochs, and every run keeps the weights of its best monitored epoch [28, Alg. 7.1]. The SNN-AE and the ANN-AE are both initialized with Xavier [29]. Left to their builders, the two arms had different defaults (fan-in scaling [30] for the SNN), and Section V shows that the initializer decides the EEG comparison, so it is fixed and shared. Within a fold the selected architecture is retrained on training ∪ validation groups, early-stopped on a recording-disjoint slice of the training groups, and evaluated once on the test group. Each fold is repeated 5 times (seeds 42–46), each repeat with its own search.

G. Energy

E is the energy of one window's forward pass, after training. It must count what spiking adds as well as what it removes, so it is the analytical model of Lemaire et al. [11], applied unchanged to every model:

$$E = E_{\text{ops}} + E_{\text{addr}} + E_{\text{mem}}, \tag{3}$$

the arithmetic, the integer additions that compute memory addresses, and the memory accesses, with the counts of Table I. The prices are Horowitz's for 45 nm at 0.9 V [6]: 32-bit float add 0.9 pJ and multiply 3.7 pJ (a MAC 4.6 pJ), 32-bit integer add 0.1 pJ, and one SRAM access 10, 20 and 100 pJ at 8 KB, 32 KB and 1 MB. As in [11], each layer holds its weights, its potentials and its input and output buffers in local SRAMs, each priced at its own size on the straight lines through those three points, and one access moves one 32-bit word. The SNN-AE's counts come from its forward pass on each test window, with that window's spikes; the ANN-AE's follow from its architecture.

Where the model leaves a choice, we take the one that works against the conclusion it could support. Memories below 8 KB, the SNN's potentials among them, are priced at 8 KB's 10 pJ. The LIF's threshold test costs one add per neuron and step, which [11] do not count. The spike encoder is charged to the SNN-AE (Poisson: a pseudo-random draw and a comparison per sample and step; latency: one spike time per sample and a comparison per sample and step); the ANN-AE has none. Leaky ReLU costs a comparison and a multiply. Reading the window from off-chip memory is left out for every model, since all start from the same buffer. Direct coding presents the same window at every step, so the SNN-AE's first layer (and the smoothing filter of the conv1d input transform in front of it) computes the same current T times; E counts it once per window, plus one write of that current and its read at every step, as an accelerator that reuses a constant input would. This choice favours the SNN-AE. Because estimates of this kind turn on the memory model [7], [17], every energy comparison is repeated with every access priced at one SRAM size (8 KB, 32 KB, 1 MB), with arithmetic alone ($4.6\,\text{pJ} \cdot N_{\text{MAC}} + 0.9\,\text{pJ} \cdot N_{\text{AC}}$, the convention of [16]), and with that first layer recomputed at every step, as our implementation runs it; Table V reports all of them. E estimates a modeled accelerator; it is not a measurement on a chip.

Table I
COUNTS PER LINEAR LAYER $n_{\text{in}} \to n_{\text{out}}$ OVER R ROWS ($R = T$ STEPS FOR THE SNN-AE, BUT 1 FOR THE FIRST LAYER OF A DIRECT-CODED INPUT; 1 FOR THE ANN-AE), AFTER [11]; θ: INPUT SPIKES.

	Analog input	Spike input
MAC	$R\,n_{\text{in}}n_{\text{out}}$	–
Add (bias, synapse)	$R\,n_{\text{out}}$	$(\theta + R)\,n_{\text{out}}$
Address add (integer)	$R(n_{\text{in}} + n_{\text{out}})$	$\theta\,n_{\text{out}}$
Read input	$R\,n_{\text{in}}$	θ
Read weight, bias	$R(n_{\text{in}} + 1)\,n_{\text{out}}$	$(\theta + R)\,n_{\text{out}}$
Read, write potential	–	$(\theta + R)\,n_{\text{out}}$ each
Write output	$R\,n_{\text{out}}$	–

LIF, per neuron and step: the leak (MAC) and the threshold test (add), and after an analog-input layer a read and a write of the potential; per output spike: the reset (add) and one write of the event. A layer that feeds an LIF writes no output of its own. Mean over T: one add per input spike, one multiply per output.

Table II
DATASETS. G: GROUP UNIT; CAP: WINDOWS KEPT PER RECORDING; d: LATENT WIDTH.

Dataset	Signal	G	#G	cap	d
FSDD	speech, 8000 Hz	speaker	6	–	16
AudioMNIST	speech, 8000 Hz	speaker	60	2	16
EEGMMIDB	EEG, 160 Hz	subject	109	40	64
Siena	EEG, 512 Hz	subject	14	40	64

IV. EXPERIMENTAL SETUP

Datasets (Table II). FSDD [31] and AudioMNIST are spoken digits at 8 kHz; EEGMMIDB [32], [33] (motor movement/imagery) and Siena [34] (epilepsy) are scalp EEG, read from the first channel of each recording (Fc5 and Fp1 in the file headers). The group is the speaker or subject. Windows do not overlap; long recordings are capped to a number of windows, and each fold keeps at most 1000 training, 150 validation and 1500 test windows, sampled round-robin across recordings.

Nested leave-one-group-out. Groups are sorted and split into 6 contiguous blocks. In fold f block f is the test set,

Table III
DESIGN STUDY, DEVELOPMENT FOLD, TEST MSE (MEAN OF 3 SEEDS; −: NOT RUN). KAIMING / XAVIER: THE INITIALIZER OF EVERY VARIANT IN THE COLUMN.

Variant	FSDD		EEGMMIDB	
	Kaiming	Xavier	Kaiming	Xavier
ANN, leaky ReLU (control)	0.667	0.654	0.489	0.293
ANN, sigmoid	0.685	–	0.308	–
ANN, clamp to $[0, 1]$	0.743	–	0.547	–
SNN, per-step loss, LIF dec., T=16	0.780	–	0.592	0.589
SNN, loss on time mean, T=16	0.734	–	0.454	–
SNN, latent mean, LReLU dec., T=4	0.724	0.729	0.484	0.449
SNN, latent mean, LReLU dec., T=16	0.729	–	0.457	0.396

block $f+1$ (mod 6) the validation set, and the rest train; for FSDD this is leave-one-speaker-out. Every window is assigned by its group before any pooling, and the harness aborts if a recording or group appears in two splits. All selection uses the validation block only.

Development fold. The design study (Section V) compared variants on fold 0 of FSDD and of EEGMMIDB, including their test windows. Those folds are therefore excluded from every headline number: FSDD and EEGMMIDB are reported on folds 1–5, AudioMNIST and Siena on folds 0–5. Including them is reported as a sensitivity analysis (Table IX).

Statistics. Every model reconstructs the same test windows. The primary estimand is the recording-level paired difference $d_r = E_{m,r} - E_{b,r}$, the SNN-AE's mean window error on recording r minus a reference model's: we report $\overline{d_r}$, a 95% bootstrap interval over recordings, and the Wilcoxon signed-rank p, Holm-corrected across the three references [35]. Group-level and seed-level tests are reported as robustness only; window-level statistics are pseudoreplicated and descriptive.

Energy first. The application is a low-power device, so we fixed in advance how the two axes are read. Where the recording-level test does not separate the SNN-AE from a reference (interval including zero or Holm-corrected $p \geq 0.05$), the model with the lower E is the one we recommend; where it does, we report what the lower-energy model gives up – its error difference against its energy ratio – instead of ranking on either axis alone. Selection inside the search is unchanged by this rule (Section III-E).

V. DESIGN STUDY ON THE DEVELOPMENT FOLD

Before the main run we asked where the gap between an earlier SNN-AE design and the matched control came from, varying one factor at a time on the development fold (1000 training windows, the search's most frequent winner, three seeds). The variants were implemented in a PyTorch replica of the network, validated against the C++ harness (EEGMMIDB: earlier SNN-AE 0.592 vs. 0.593; FSDD: ANN-AE 0.667 vs. 0.662). Table III lists the test MSE.

Three findings shaped the method. *(i) Two costs were not the spike.* Scoring the reconstruction of every simulation step asks each step, carrying at most one binary event per neuron, to reproduce the analog window alone; scoring the time mean removed 0.046 on FSDD. A spiking decoder added error on top of the encoder's; averaging the latent and decoding once with leaky ReLUs is as good and runs the decoder once instead of T times. *(ii) The initializer decided the EEG comparison.* With fan-in scaling in both arms the control stopped at 0.489 (0.479 with a 200-epoch cap, so not for lack of epochs) and the SNN-AE looked equal to it (0.484); with Xavier in both arms the control reaches 0.293 and the SNN-AE 0.449. Xavier is the better or equal choice for each arm on both datasets, so the matched comparison is also a best-against-best one. *(iii) What remains is saturation.* A non-spiking network with a hard clamp lands among the spiking variants (FSDD 0.743, against 0.734 for the SNN trained on the time mean; EEG 0.547, against 0.592 for the per-step SNN), while a sigmoid, bounded but smooth, stays near the control (FSDD 0.685 vs. 0.667; EEG 0.308, near the Xavier control's 0.293). A reason, argued rather than measured: under direct input an LIF neuron sees the same current I at every step, so with a hard reset it never fires for $I < V_{th}(1 - \beta)$ and fires at every step for $I \geq V_{th}$ – a clipped transfer function.

T mattered on EEG (0.449 at T=4, 0.396 at T=16) but not on FSDD (0.717 to 0.729 for T from 1 to 16), which is why T is searched rather than fixed. Every choice above came out the same on the fold's validation windows (e.g. EEG, Xavier: 0.425 vs. 0.372 for $T = 4$ vs. 16), but the test windows were seen, hence the development-fold rule of Section IV. A check of the adopted design in the C++ harness on the same fold (two runs, small search) moved the SNN-AE from 0.827 to 0.768 on FSDD and from 0.593 to 0.532 on EEG, against controls of 0.652 and 0.302.

A. Pre-validation of sequence baselines

The submitted version also searched LSTM, GRU and Transformer autoencoders. They differ from the SNN-AE in recurrence, sequence length and capacity at once, so they cannot isolate what the spike costs, and their searches dominated the grid's compute. Before leaving them out we measured them once, on the development folds (FSDD fold 0 and EEGMMIDB fold 0). LSTM-AE and GRU-AE stack $L \in [1, 3]$ recurrent layers of width $H \in [8, 256]$ (floored at d), project the last state to d and decode by repeating a projection of z over time. The Transformer-AE embeds frames to $d_{\text{model}} \in [16, 128]$, adds sinusoidal positions, applies $N \in [1, 4]$ self-attention blocks (heads $\in \{1, 2, 4, 8\}$, feed-forward width $\in [32, 256]$), mean-pools to d and decodes with N further blocks and no cross-attention, so all information passes through z. All three read the window as $S = 32$ frames of 8 samples, so only the direct code is defined for them. Every family, the SNN-AE included, ran one NSGA-II of population 4 for 2 generations (12 evaluations, front re-scored over 2 seeds), 2 runs (seeds 42–43), with the windows, training and selection of Sections III-E–III-F, each family in its own process on one CPU thread. Their E applies the counts of Table I to every matrix product of a cell or an attention block at each of the S positions; gate, softmax and normalization nonlinearities cost an add and a multiply each, below an exact exponential, which favours them.

Table IV

PRE-VALIDATION, DEVELOPMENT FOLDS, 2 RUNS. FOLD: WALL TIME OF THE FAMILY'S FOLD, DATA LOADING EXCLUDED (SEARCH, RE-SCORING, FINAL FIT, TEST); ×SNN: ITS RATIO TO THE SNN-AE'S, WHOSE FOLD ALSO TRAINS THE ANN-AE. TRAIN: FINAL FIT, PER TRAINING WINDOW AND EPOCH. INFER.: ONE WINDOW RECONSTRUCTED AFTER TRAINING. ONE CPU THREAD EACH.

Model	Test MSE	Fold (h)	×SNN	Train (ms/win.)	Infer. (ms/win.)	E (nJ)
FSDD						
SNN-AE	–	–	–	–	–	–
ANN-AE	–	–	–	–	–	–
LSTM-AE	–	–	–	–	–	–
GRU-AE	–	–	–	–	–	–
Transformer-AE	–	–	–	–	–	–
PCA ($k = d$)	–	–	–	–	–	–
EEGMMIDB						
SNN-AE	–	–	–	–	–	–
ANN-AE	–	–	–	–	–	–
LSTM-AE	–	–	–	–	–	–
GRU-AE	–	–	–	–	–	–
Transformer-AE	–	–	–	–	–	–
PCA ($k = d$)	–	–	–	–	–	–

Table IV reports the cost of training (the fold's wall time) and of processing data after training (the measured time to reconstruct one window, and E): [the pre-validation has not run yet]. On quality, [the pre-validation has not run yet]. The reduced budget can change any family's winner, and with it both its error and its per-window cost; the table describes the winners found here, not the families' best.

A Transformer whose layer normalization follows each residual sum, as ours does, can need a learning-rate warm-up to train at a fixed Adam rate; one that normalizes before each sub-layer does not [36]. We therefore retrained the Transformer-AE winners of both development folds, 2 seeds each, with a linear warm-up over 4000 steps and, separately, with pre-norm blocks closed by a final layer normalization: on FSDD fold 0, 1 of 8 warm-up trainings and 1 of 8 pre-norm trainings left the mean-predictor plateau (control: 0 of 8), and the mean test MSE of the final fits was 0.916 with warm-up, 0.931 with pre-norm and 0.935 for the control, against 0.996 for the mean predictor (best single fit: 0.867, warm-up); on EEGMMIDB fold 0, 0 of 8 warm-up trainings and 1 of 8 pre-norm trainings left the mean-predictor plateau (control: 1 of 8), and the mean test MSE of the final fits was 0.993 with warm-up, 0.921 with pre-norm and 0.992 for the control, against 1.001 for the mean predictor (best single fit: 0.792, pre-norm).

VI. RESULTS

Draft: the tables of this section are placeholders (−) until the nested-LOSO grid has run; they are regenerated from the harness's outputs.

Table VI gives the test MSE and energy of every model on the headline folds, Table VII the recording-level differences between the SNN-AE and each reference, Fig. 2 the same models in the quality–energy plane, Table V the energy under each repricing, and Table VIII what the SNN-AE's search selected. On energy per window, [the energy comparison has not run yet].

Table V

ENERGY PER WINDOW (NJ), HEADLINE FOLDS, MEAN OVER EVERY WINNER: E (SECTION III-G), ITS SHARE SPENT ON MEMORY ACCESSES, E WITH EVERY ACCESS PRICED AT ONE SRAM SIZE, ARITHMETIC ALONE, AND E AS EXECUTED (A DIRECT-CODED FIRST LAYER RECOMPUTED AT EVERY STEP). LAST ROW OF EACH DATASET: THE SNN-AE'S ENERGY OVER THE ANN-AE'S, PER COLUMN.

Model	E (nJ)	Mem. share	E, one SRAM size 8 KB	32 KB	1 MB	Arith. only	As exec.
FSDD							
SNN-AE	–	–	–	–	–	–	–
ANN-AE	–	–	–	–	–	–	–
SNN-AE ÷ ANN-AE	–	–	–	–	–	–	–
AudioMNIST							
SNN-AE	–	–	–	–	–	–	–
ANN-AE	–	–	–	–	–	–	–
SNN-AE ÷ ANN-AE	–	–	–	–	–	–	–
EEGMMIDB							
SNN-AE	–	–	–	–	–	–	–
ANN-AE	–	–	–	–	–	–	–
SNN-AE ÷ ANN-AE	–	–	–	–	–	–	–
Siena							
SNN-AE	–	–	–	–	–	–	–
ANN-AE	–	–	–	–	–	–	–
SNN-AE ÷ ANN-AE	–	–	–	–	–	–	–

Reading guide. The first question is the gap between the SNN-AE and its matched control, per dataset and modality, and whether its interval excludes zero. The second is whether the SNN-AE saves energy for it: its E against the ANN-AE's, and whether that ordering survives every column of Table V, read together with the selected encoding and T in Table VIII: binary inputs turn the first layer's MACs into adds, but every step adds a read and a write of every potential. The third places both against PCA; a model that does not beat PCA has not shown that its nonlinearity helps on that dataset. Orderings are asserted only where the Holm-corrected p is below 0.05 and the interval excludes zero, and a finding that holds on one modality only is reported as such.

VII. DISCUSSION

The design study already constrains what the main run can show. The spike's cost in this autoencoder is not a property of discrete events as such: a non-spiking network with the same clipped transfer function pays it, and a smooth bounded one does not. Remedies that act on the transfer function – a membrane read out instead of spikes in some layers, adaptive thresholds, or soft resets, which made things worse here – are therefore the levers to test. Longer simulations helped on EEG only (Table III) and cost arithmetic in proportion to T and in memory traffic. On energy, binary inputs save the first layer's multiplies while every step pays for the potentials, so whether the SNN-AE ends up cheaper than its control depends on the encoding, T and spike rate the search selects, not on spiking alone.

Limitations. (i) *Groups.* Recordings are nested in 6, 60, 109 and 14 groups (5 FSDD speakers on the headline folds). The recording-level test treats recordings as independent, so it speaks to new recordings of these speakers and subjects,

Table VI

TEST MSE (MEAN ± STD OVER 5 SEEDS; LOWEST PER DATASET IN BOLD) AND ENERGY PER WINDOW E (NJ, MEMORY-AWARE, SECTION III-G) ON THE HEADLINE FOLDS. PCA AND MEAN ARE FITTED ONCE (NO SEEDS, NO STD) AND ARE NOT COSTED.

Model	FSDD MSE	E	AudioMNIST MSE	E	EEGMMIDB MSE	E	Siena MSE	E
Mean	–	–	–	–	–	–	–	–
PCA	–	–	–	–	–	–	–	–
SNN-AE	–	–	–	–	–	–	–	–
ANN-AE (matched)	–	–	–	–	–	–	–	–

Table VII

SNN-AE MINUS REFERENCE, RECORDING LEVEL, HEADLINE FOLDS: $\overline{d_r}$ (POSITIVE: THE SNN-AE ERRS MORE; *: HOLM-CORRECTED WILCOXON $p < 0.05$) AND ITS 95% BOOTSTRAP INTERVAL.

Dataset	n_{rec}	ANN-AE	PCA	Mean
FSDD	–	–	–	–
AudioMNIST	–	–	–	–
EEGMMIDB	–	–	–	–
Siena	–	–	–	–

Quality–energy plane: generated from data/paper_loso_⟨dataset⟩_pareto.csv after the grid run.

Figure 2. Test MSE (mean ± std over seeds) against energy per window E, headline folds. Down and left is better; the dashed line is PCA with $k = d$.

Table VIII

SNN-AE SELECTED PER DATASET OVER EVERY (FOLD, RUN): MODAL ENCODING AND INPUT TRANSFORM (k/n AGREEING, IF NOT ALL), MEDIAN T [RANGE], MEDIAN INITIAL THRESHOLD AND DECAY, MEDIAN DEPTH L, AND THE HIDDEN NEURONS' SPIKE FRACTION ρ ON TEST WINDOWS.

Dataset	Enc.	Transf.	T	$V_{th}^{(0)}$	$\beta^{(0)}$	L	ρ
FSDD	–	–	–	–	–	–	–
AudioMNIST	–	–	–	–	–	–	–
EEGMMIDB	–	–	–	–	–	–	–
Siena	–	–	–	–	–	–	–

Table IX

SENSITIVITY TO THE DEVELOPMENT FOLD: SNN-AE MINUS ANN-AE, RECORDING LEVEL, WITHOUT (HEADLINE) AND WITH IT.

Dataset	Folds	n_{rec}	$\overline{d_r}$	95% CI	p_{Holm}
FSDD	–	–	–	–	–
	–	–	–	–	–
EEGMMIDB	–	–	–	–	–
	–	–	–	–	–

not to new ones; the group-level tests that would are reported as robustness only, and with 5 speakers the exact Wilcoxon p cannot fall below 0.0625. (ii) *One channel, short windows.* EEG is read from one channel and every window has 256 samples; results may not transfer to multichannel or longer contexts. (iii) *Estimated, not measured energy.* E models an accelerator with local SRAMs at 45 nm and leaves out off-chip traffic, leakage and control; a real chip's consumption, and whether it can exploit sparsity, decide the final number. The repricings bound how much the conclusion depends on the memory model, not on the chip. (iv) *Search budget.* The SNN-AE evaluates 48 candidates per fold and run, and the control none (it inherits the SNN-AE's widths), which favours the SNN-AE over its control; a larger budget could change the winner. (v) *Matched, not identical inputs.* The ANN-AE always reads the analog window, whereas the SNN-AE may read spikes; the information available is the same, its code is not. (vi) *Fixed latent width.* d follows prior autoencoder practice for each domain, not a sweep on these datasets. (vii) *Training cap.* At most 30 epochs; on the development fold most EEG runs were still improving at the cap. (viii) *Sequence models pre-validated, not compared.* The LSTM-, GRU- and Transformer-AE were searched with 12 evaluations on two development folds only, so the conclusions concern spiking against its matched control, not against sequence models.

VIII. Conclusion

We compared a spiking autoencoder with a matched non-spiking control and PCA on two speech and two EEG datasets, under nested leave-one-group-out cross-validation with every choice made on validation groups and the development fold kept out of the headline numbers; searched sequence autoencoders were measured in a pre-validation on the same folds for what they cost to train and to run. Matching the control exposed two confounds that would have changed the conclusion – a per-step loss and a spiking decoder that inflated the gap, and an initializer that, shared but poorly chosen, made it vanish on EEG – and located the remaining cost in the saturation of the LIF transfer function. The harness, profile and analysis scripts are released [37].

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